RAMDEOBABA UNIVERSITY, NAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech. CSE (Data Science and Analytics) (DSA)

SEMESTER V

LAB

PROJECT REPORT

CYBERCRIME AND THREAT INTELLIGENCE ANALYTICS DASHBOARD

DST Lab Project Report

Submitted by

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1. PROJECT OBJECTIVE

The objective of this project is to develop an interactive Tableau dashboard for descriptive analysis of cybercrime incidents and threat-intelligence attributes. The workbook organizes incident information into three dashboards: Global Cybercrime Overview, Threat Intelligence Analysis, and Financial Impact and Incident Response. It brings together event counts, geographic coverage, attack categories, affected users, financial loss, defense mechanisms, vulnerability types and incident-resolution time in a visual reporting environment.

The project aims to:

- Load and inspect the cybercrime and threat-intelligence dataset and understand its available fields.
- Summarize recorded events by attack type, target industry and year.
- Explore the geographic distribution of recorded cyber threats and the number of countries represented.
- Compare threat source categories, vulnerability types and defense mechanisms.
- Present summary measures for total events, affected users and financial loss, together with resolution-time analysis.
- Arrange related worksheets into three dashboards to support visual exploration and communication.

2. INTRODUCTION

Cybercrime includes malicious activities carried out through digital systems, networks and connected technologies. Incidents can affect individuals, businesses, public institutions and critical services. Understanding the patterns of recorded incidents is important for security awareness, prioritization and the communication of cyber-risk information. Incident datasets may contain multiple dimensions, such as country, region, year, attack type, target industry, attack source, security vulnerability and defense mechanism, alongside impact measures.

This project, “CyberCrime and Threat Intelligence Analytics Dashboard,” uses Tableau to organize these fields into interactive visual summaries. Rather than presenting rows of incident records in isolation, the workbook groups related views so that users can explore incident distribution, threat characteristics and impact-related measures. The dashboard is intended for descriptive analytics; it does not claim to identify an attacker, predict future attacks or establish causation.

2.1 Problem Statement

Cybercrime incident data can be difficult to interpret when its categorical fields and impact measures are viewed only as a flat table. A user may need to compare attack types, target industries, geographic coverage, threat sources, vulnerabilities, defense mechanisms and resolution time separately. The project addresses this presentation challenge by creating a structured set of Tableau worksheets and combining them into three dashboards. The resulting views support comparison and exploration while keeping the interpretation limited to the records and measures available in the dataset.

3. PROJECT METHODOLOGY

The project workflow consists of connecting the dataset to Tableau, inspecting the available fields, preparing the data for analysis, creating calculated fields, developing individual worksheets, arranging worksheets into dashboards and checking the workbook structure.

3.1 Dataset Loading and Exploration

The Tableau workbook identifies the data source as **Cybercrime_Threat_Intelligence_Sample_Dataset**, with the underlying file referenced as **Cybercrime_Threat_Intelligence_Sample_Dataset.csv**. The CSV itself was not supplied separately with the workbook in this task. Therefore, the exact record count, field-level missing-value counts, duplicate count and numerical summary statistics could not be independently calculated for this report.

The workbook schema exposes the following source fields:

Field	Role in the analysis
Incident_ID	Identifier for a recorded incident
Country, Region	Geographic dimensions
Year	Time dimension
Attack Type	Incident classification
Target Industry	Industry category affected
Financial Loss (in Million $)	Financial impact measure
Number of Affected Users	Affected-user measure
Attack Source	Source category
Security Vulnerability Type	Vulnerability classification
Defense Mechanism Used	Defense/control category
Incident Resolution Time (in Hours)	Resolution-time measure
Data_Status	Data-status field present in the source schema

The workbook schema confirms the listed fields and their use in the worksheet titles and calculations. It does not, by itself, verify the actual distribution of values or the completeness of the CSV data.

3.2 Data Preprocessing

- Review field names and data types in Tableau and confirm that identifiers, categorical dimensions, year and numerical measures are interpreted appropriately.
- Check categorical values for inconsistent spelling, spacing or capitalization before comparing categories.
- Inspect the Data_Status field and review records requiring attention; the workbook alone does not establish the number of affected records.
- Check for null or blank values and duplicate incident identifiers in the source CSV before final publication. No counts are asserted here because the CSV was not provided separately.
- Confirm that financial loss and resolution time are numeric measures and that year is treated as a time-related dimension.
- Use consistent labels and units across worksheet titles, axes, tooltips and dashboard captions.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version / status	Purpose
Tableau Desktop / Tableau workbook	Version not stated in the report materials	Connect to the dataset, create worksheets, calculated fields and dashboards.
CSV dataset	Cybercrime_Threat_Intelligence_Sample_Dataset.csv; source version not supplied	Provide tabular cybercrime incident records to Tableau.
Tableau calculated fields	Not applicable	Aggregate financial loss, affected users and distinct countries.
GitHub	Repository link supplied by the team	Host or share project files and related documentation.

4.2 Step-wise Implementation

1. Connect Tableau to the cybercrime CSV data source and confirm that the expected source fields are available in the Data pane.
2. Inspect the field types and distinguish dimensions such as Country, Region, Attack Type and Target Industry from numerical measures such as Financial Loss, Number of Affected Users and Incident Resolution Time.
3. Create the three aggregate calculated fields used in the workbook: Total Financial Loss, Total Affected Users and Countries Represented.
4. Develop the event overview worksheets, including Recorded Events by Attack Type, Recorded Events by Target Industry and Recorded Threat Events by Year.
5. Develop geographic and KPI worksheets, including Geographic Distribution of Recorded Cyber Threats, Countries Represented, KPI - Total Events, KPI - Affected Users and KPI - Financial Loss.
6. Develop threat-intelligence worksheets for Recorded Threats by Defense Mechanism, Recorded Threats by Source Category and Recorded Threats by Vulnerability Type.
7. Develop impact and response worksheets for Financial Loss by Attack Type and Average Recorded Resolution Time by Attack Type.
8. Assemble the worksheets into Global Cybercrime Overview, Threat Intelligence Analysis and Financial Impact and Incident Response.
9. Review titles, labels, number formats, legends and dashboard layout. Validate displayed values against the source data before final submission.

5. TABLEAU CALCULATED FIELDS

The project is implemented in Tableau, so the workbook uses Tableau calculated fields rather than DAX measures. The following formulas are present in the supplied workbook:

5.1 Total Financial Loss

```
SUM([Financial Loss (in Million $)])
```

Aggregates the financial-loss measure across the records in the current view and filter context.

5.2 Total Affected Users

```
SUM([Number of Affected Users])
```

Aggregates the number of affected users across the records in the current view and filter context.

5.3 Countries Represented

```
COUNTD([Country])
```

Counts distinct country values represented in the current view and filter context.

The calculated fields provide aggregate measures for the dashboards. Their displayed values can change according to the dimensions and filters used in a worksheet. Since the CSV data was not separately supplied, this report does not state numerical KPI results.

5.4 Worksheet Inventory

1. Average Recorded Resolution Time by Attack Type
2. Countries Represented
3. Financial Loss by Attack Type
4. Geographic Distribution of Recorded Cyber Threats
5. KPI - Affected Users
6. KPI - Financial Loss
7. KPI - Total Events
8. Recorded Events by Attack Type
9. Recorded Events by Target Industry
10. Recorded Threat Events by Year
11. Recorded Threats by Defense Mechanism
12. Recorded Threats by Source Category
13. Recorded Threats by Vulnerability Type

6. RESULTS AND DISCUSSION

The supplied Tableau workbook contains three dashboards and thirteen worksheets. The workbook file stores dashboard and worksheet definitions, but it does not include rendered dashboard screenshots. The following figures are therefore clearly identified as structural illustrations derived from the workbook's dashboard and worksheet names. They are not screenshots of the live Tableau interface and do not display invented data values.

6.1 Global Cybercrime Overview



Figure 1. Structural illustration of the Global Cybercrime Overview dashboard based on the supplied Tableau workbook.

This dashboard groups the high-level incident measures and overview worksheets. Its workbook structure includes KPI - Total Events, KPI - Affected Users, Countries Represented, Recorded Events by Attack Type, Recorded Events by Target Industry, Recorded Threat Events by Year and Geographic Distribution of Recorded Cyber Threats. These views are intended to provide a broad summary of the recorded incident dataset. Actual KPI values and category distributions must be read from the connected Tableau data source.

6.2 Threat Intelligence Analysis

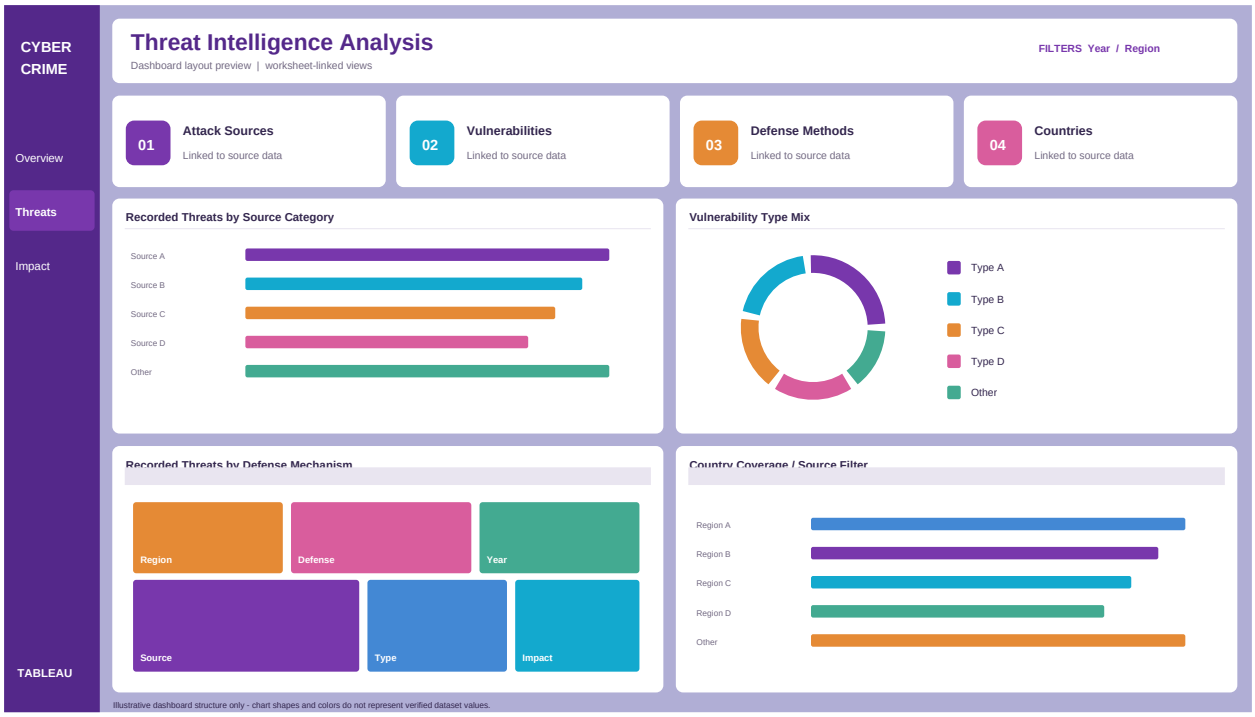


Figure 2. Structural illustration of the Threat Intelligence Analysis dashboard based on the supplied Tableau workbook.

This dashboard organizes the threat-characteristic worksheets. Recorded Threats by Source Category groups incidents by the source classification; Recorded Threats by Vulnerability Type organizes them by the recorded vulnerability category; and Recorded Threats by Defense Mechanism groups them by the defense mechanism used. Countries Represented is also available as a distinct-country summary worksheet. The dashboard supports comparison of recorded threat attributes; it does not independently verify attribution or the effectiveness of a defense mechanism.

6.3 Financial Impact and Incident Response

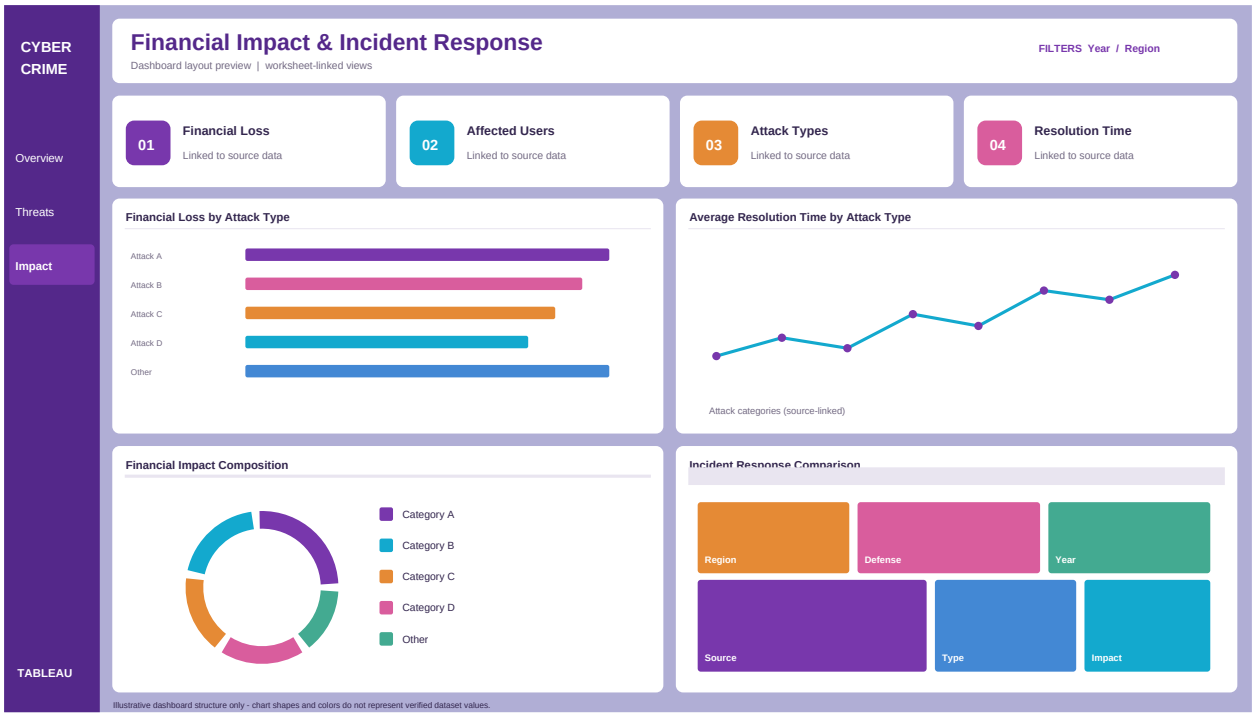


Figure 3. Structural illustration of the Financial Impact and Incident Response dashboard based on the supplied Tableau workbook.

This dashboard focuses on impact and response measures. Financial Loss by Attack Type compares the recorded financial-loss measure across attack categories. Average Recorded Resolution Time by Attack Type compares the resolution-time measure across attack categories. KPI - Financial Loss and KPI - Affected Users provide headline impact measures. The exact numerical results are not reproduced here because the CSV data and rendered Tableau values were not supplied.

6.4 Discussion and Interpretation

The three-dashboard structure separates the analysis into an overview of incident distribution, a closer examination of threat characteristics, and a view of financial impact and incident response. This arrangement makes the workbook easier to navigate and helps users focus on related questions. Interpretation should remain descriptive: totals depend on the records and filters included in the view, and aggregate financial loss or affected-user counts should not be treated as independently verified external estimates. The workbook alone does not establish causation, attacker identity, future attack probability or the effectiveness of security controls.

7. CONCLUSION

The project presents a Tableau-based cybercrime and threat-intelligence analytics workbook organized into three dashboards: Global Cybercrime Overview, Threat Intelligence Analysis, and Financial Impact and Incident Response. It contains thirteen worksheets covering event counts, attack types, target industries, year-wise records, geographic distribution, threat sources, vulnerability types, defense mechanisms, financial loss, affected users and incident-resolution time. Three calculated fields aggregate total financial loss, total affected users and distinct countries.

The workbook provides a structured visual approach for exploring cybercrime incident records and communicating descriptive patterns. The main limitation of this report is that the CSV source file and rendered dashboard screenshots were not included with the uploaded workbook. Consequently, dataset statistics, KPI values, missing-value counts and detailed analytical findings have not been invented. Before final academic submission, the team should validate the workbook against the original CSV, review data quality, and replace the structural illustrations in this report with exported screenshots from Tableau if screenshots are specifically required by the evaluator.

8. GITHUB LINK

https://github.com/Trupti-Avhadkar/CyberCrime-Threat-Intelligence-Analytics

9. REFERENCES

[1] Tableau, “Tableau Help.” https://help.tableau.com/

[2] Tableau, “Create a calculated field.” https://help.tableau.com/current/pro/desktop/en-us/calculations_calculatedfields_create.htm

[3] Tableau, “Build a dashboard.” https://help.tableau.com/current/pro/desktop/en-us/dashboards.htm

[4] Cybercrime_Threat_Intelligence_Sample_Dataset.csv, dataset referenced by the supplied Tableau workbook. The original dataset download/source URL was not included in the supplied materials.

[5] Ramdeobaba University, Department of Computer Science and Engineering, “DST Lab Project Report Format,” Semester V, Academic Year 2026–27 (reference PDF supplied for formatting).